FSD CIA 2: OpenStreetMap UX Improvement

Beacon — Find Nearby Essentials (React, Tailwind CSS, JavaScript)

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Part I – Problem Definition and Innovation

1. Benchmark Analysis

I studied the search, place details and directions workflow on openstreetmap.org. My target users are people new to a city, like students who just moved to Bengaluru and need to find a pharmacy, ATM, hospital or washroom quickly.

OSM already has a search box, a details page for every place, "Query features", directions and map layers. While testing it I found these problems:

- "pharmacy near MG Road, Bengaluru" gives long address lines with no distance, no open/closed status and no nearest-first order.
- "pharmacy near me", with the map on Bengaluru, returned pharmacies in Italy and Venezuela.
- From an India-wide view, "pharmacy near MG Road" picked MG Roads in West Bengal and Mumbai, and never asked which one.
- Opening hours are shown as raw tags like PH,Mo-Su 08:00-23:30.
- A place page only has Download XML and History, no directions or call button.
- Query features lists waste baskets and footpaths along with real places.

Around MG Road only 17 out of 233 places had opening hours on OSM, so the app must say clearly when hours are not known.

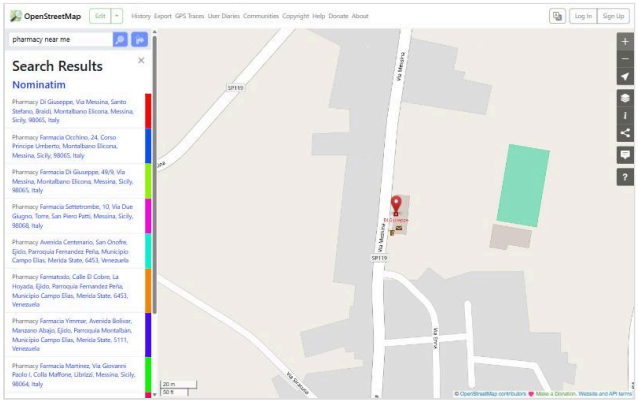


Figure 1: "pharmacy near me" opens Italy

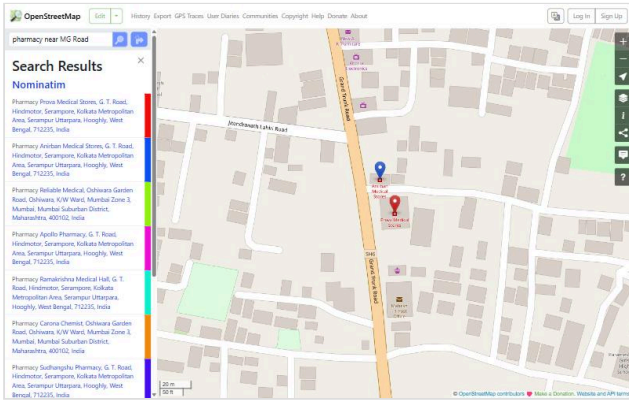


Figure 2: MG Road from the wrong city

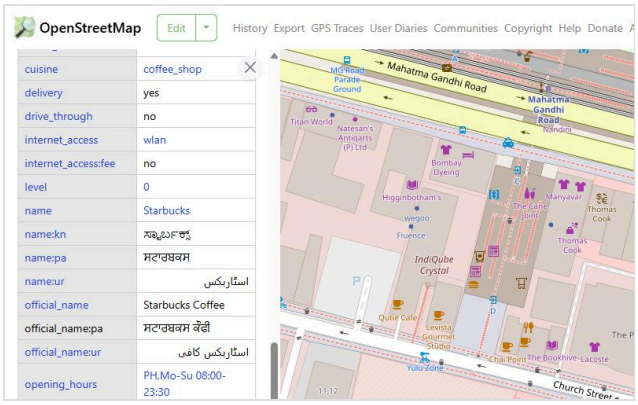


Figure 3: raw opening hours tag

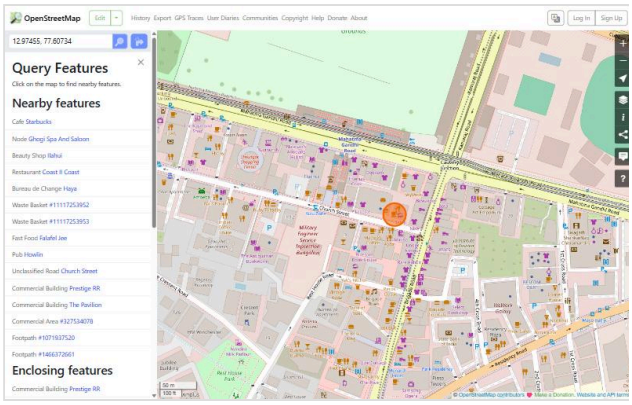


Figure 4: waste baskets in Query features

2. Problem Definition

A newcomer cannot quickly find the nearest essential place that is open right now on OpenStreetMap. "near me" ignores their location, street names land in the wrong city, results have no distance or open status, and there is no simple way to get directions. In practice this means opening many results, reading raw tags, and sometimes walking to a closed shop.

3. Proposed Solution

Beacon is a focused web app on OpenStreetMap data. You type what you need the way you would say it, and it shows the nearest places with distance, walking time and whether they are open, then takes you there.

Search understands "pharmacy near me", "atm near MG Road" or "24/7 hospital nearby". "near me" uses your location, or your city if location is off. When a street name exists in more than one place it shows cards with the area, city and distance so you can pick, or it picks the nearest one if that is obvious. Each place has a card with Go Now, Call, Website, Save and Share, and Start opens turn-by-turn navigation for walking, cycling or driving.

The interface is built with HTML5 and Tailwind CSS. On phones the search sits on top with category chips under it, results come up in a bottom sheet and there is a tab bar at the bottom. On desktop the same parts sit in a left panel. It uses Apple's Liquid Glass style, a 3D globe at the start, and light theme by default.

Part II – JavaScript Implementation

4. Events

Typing in the search box (input, debounced) loads suggestions, submitting the form runs the search, clicking a pin opens the place, right-click shows the address of that spot, keydown handles shortcuts (/ search, Esc close, L locate, 1–9 categories), pointer events drag the bottom sheet, online/offline shows a banner, and the microphone uses the speech recognition result and end events.

5. Form with Validation

The Report a Problem form checks the problem type, place name, details (at least 15 characters), optional email or phone, the location pin and a consent tick. Each field is checked when you leave it and all of them on submit. Errors show under the field and as a list at the top. A valid report is saved on the device and can be posted as an OSM Note.

6. Fetch API and JSON

Suggestions come from Photon, places and nearby search from Nominatim, extra details from Overpass, routes from OSRM and place photos from Wikidata. All responses are read as JSON and converted into one common place format.

7. Errors, Loading and Empty States

Skeleton loaders show while searching and "Finding your location" while locating. Nominatim results appear first and Overpass details are added later, since Overpass took 29 to 43 seconds in my tests. Every request has a timeout with a clear message and a retry button. Empty results offer a wider search, and if location is denied the app uses the chosen city and says so.

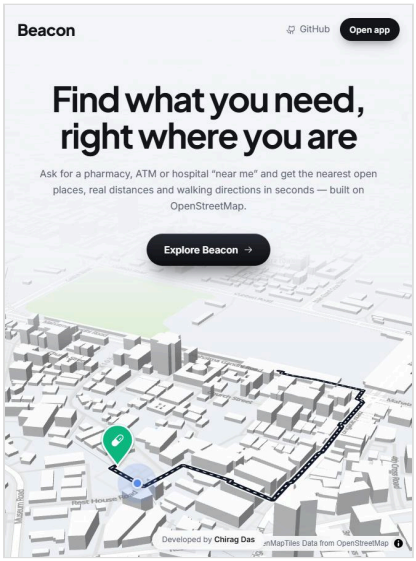


Figure 5: landing page

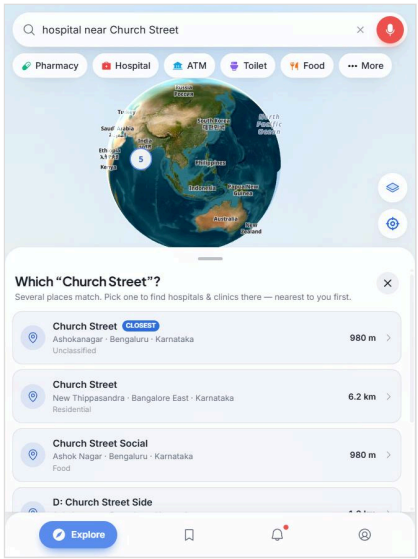


Figure 6: choosing the right Church Street

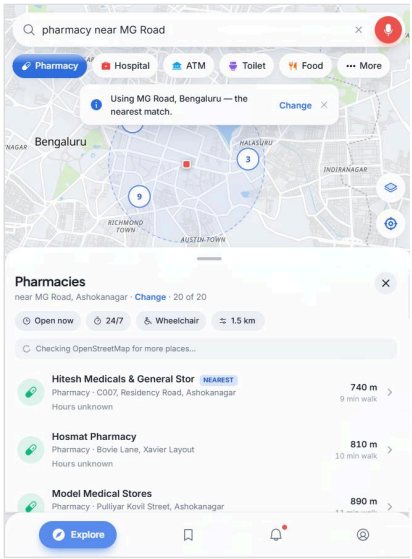


Figure 7: nearest pharmacies

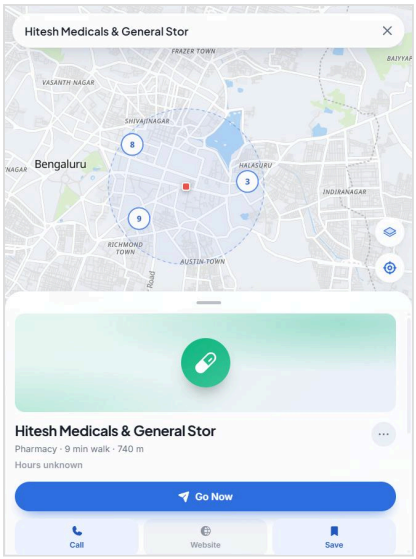


Figure 8: place card

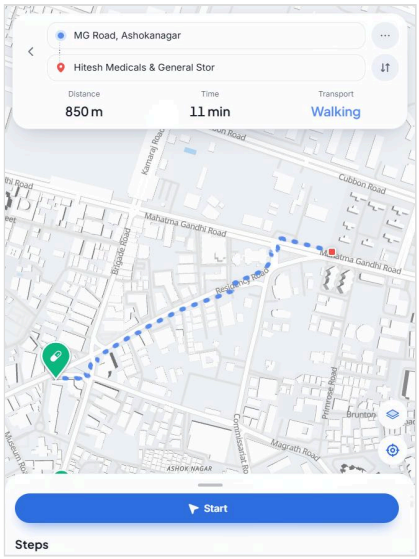


Figure 9: walking route

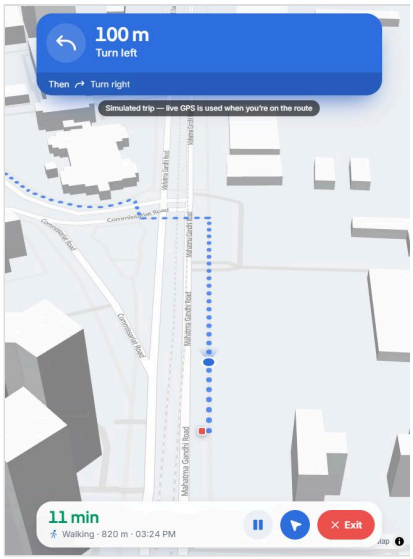


Figure 10: navigation

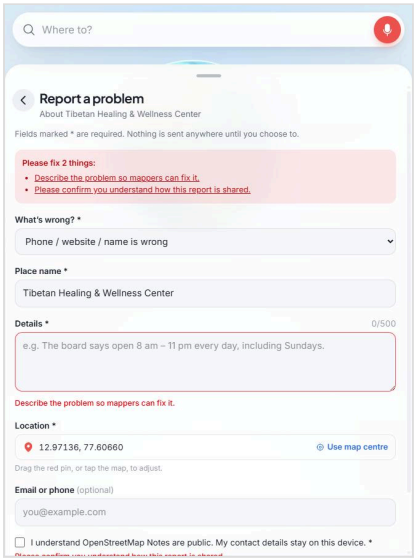


Figure 11: form validation

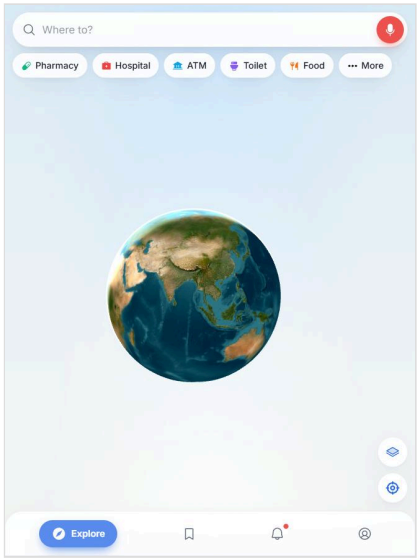


Figure 12: 3D globe

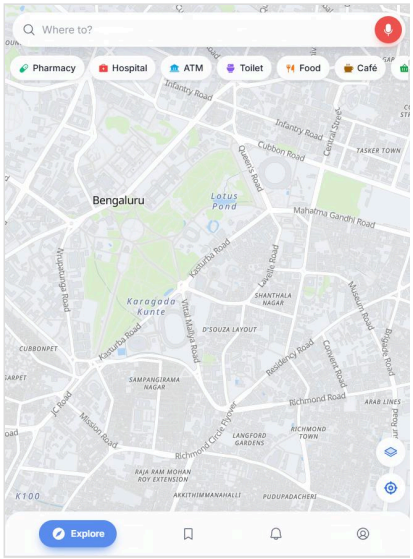


Figure 13: explore view

Part III – Documentation, GitHub and Deployment

8. Links

Live app: chiragdascodes.github.io/FSDCIA1
GitHub: github.com/Chiragdascodes/FSDCIA1

The repository has a README and commits for each stage. Every push runs the tests, builds the app and deploys it to GitHub Pages. To run it locally: npm install, then npm run dev.

9. Testing

There are 64 automated tests and all of them pass. They cover the search parser, choosing the right MG Road, opening hours, distances, form validation, fetch error handling and navigation. I also tested by hand: "pharmacy near MG Road" picks Bengaluru and lists 20 pharmacies nearest first, "hospital near Church Street" asks which of the two Church Streets I mean, location denied falls back to Bengaluru, the form only saves valid input, Go Now and Start follow the route, results still show when Overpass is down, and the layout works on phone and desktop without sideways scrolling.

10. Limitations

- Most places on OSM have no opening hours yet.
- The free Overpass and routing servers can be slow or busy.
- No live traffic or public transport routes.
- Navigation needs real GPS; on a laptop it plays a simulated trip.
- Voice search works only in Chrome, Edge and Safari.

11. Self-Learning (Beyond the Task)

The paper only asked for a responsive page with events, a form and Fetch. I went further and learned these on my own:

- 3D globe that turns into a flat street map as you zoom in, using MapLibre's globe projection.
- Apple's Human Interface Guidelines and the Liquid Glass design style.
- Google Maps style navigation: snapping GPS onto the route, live heading and a different camera for walking, cycling and driving.
- Voice search with the Web Speech API.
- Picking the right place when a name repeats, by ranking matches by distance from the user.
- Reading OSM opening hours and turning them into "Open · closes 23:30".
- Place photos and brand logos from Wikidata and Wikimedia Commons.
- Installable app (PWA) that opens offline using a service worker.
- 64 unit tests with Vitest and automatic test and deploy on every push with GitHub Actions.

- Accessibility: keyboard shortcuts, screen reader labels, reduced motion and reduced transparency support.

12. AI Usage Declaration

- I chose the problem, the target users and the workflow to improve.
- I found the wrong MG Road issue myself while testing OpenStreetMap.
- I designed the custom UI in Figma and set the look: landing page, white and blue theme, Liquid Glass, circular icons, light theme.
- I decided the features, including Google Maps style navigation and no horizontal scrolling.
- I tested every version and asked for changes until it worked the way I wanted.
- I used Claude (Anthropic) as an assistant: it helped analyse openstreetmap.org, wrote most of the code from my instructions and helped draft this report.
- I have reviewed the code and can explain how it works.